Pixel Art Asset Specification — Clawbada Lobsters

Handoff document for pixel artist + developer System: `@clawbada/asset-gen` procedural pixel art generator

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1. Overview

Clawbada lobsters are procedurally generated pixel art characters. Each lobster is composed of **6 body parts**, each drawn from one of **10 class themes**. The system generates **16 visual variants** from each hand-authored base template, yielding **960 unique body part appearances** from just **60 base templates**.

What You're Making

Item	Count	Description
Base templates	60	6 body parts × 10 classes. Hand-authored JSON pixel files.
Procedural variants	960	16 auto-generated variants per template. You define where mutations can happen.
Assembled lobsters	Billions	Any combination of 6 body parts from any class. The system composites them.

Key Numbers

Property	Value
Canvas size	**48 × 48 pixels**
Display sizes	48, 96, 192, 384 (nearest-neighbor upscale)
Colors per template	**7 palette roles** (not raw colors)
Templates needed	**60 total** (3 exist as reference, 57 remaining)
Target pixel count	**300-800 pixels** per template (typical)
File format	Sparse JSON (only non-transparent pixels)

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2. How the System Works

```
Artist creates base template (JSON)
  ↓
System generates 16 variants (mutations within defined zones)
  ↓
System applies class palette colors (template role 2 → class primary color)
  ↓
System applies breed type color shift (hue rotation, saturation adjust)
  ↓
6 body parts composited into one lobster (layered back-to-front)
  ↓
Optional evolution effects (glow, particles for evolved lobsters)
  ↓
Optional legend effects (per-class special treatment for rare legends)
  ↓
Upscale to display size (nearest-neighbor, stays crisp)
```

Critical concept: You paint with **role indices** (numbers 0-6), not actual colors. The same template renders in any of the 10 class color schemes. Role 2 is always "primary base color" — it becomes steel blue for Bulwark, jade green for Mantis, crimson for Reaver, etc.

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3. Template JSON Format

Each template is a JSON file with this exact structure:

```json
{
  "bodyPart": "carapace",
  "classAffinity": 0,
  "version": 1,
  "width": 48,
  "height": 48,
  "anchor": { "x": 24, "y": 20 },
  "bounds": { "x": 12, "y": 11, "w": 24, "h": 18 },
  "mutationZones": [
    {
      "x": 14, "y": 13, "w": 20, "h": 4,
      "allowed": ["shift_pixels", "thicken", "detail_overlay"]
    }
  ],
  "pixels": [
    {"x":18,"y":11,"role":0},
    {"x":19,"y":11,"role":0},
    {"x":20,"y":12,"role":2}
  ]
}
```

Field Reference

Field	Type	Description
`bodyPart`	string	One of: `carapace`, `claws`, `tail`, `antennae`, `eyes`, `legs`
`classAffinity`	number	Class index 0-9 (see Class Palettes)

Field	Type	Description
`version`	number	Always `1`
`width`	number	Always `48`
`height`	number	Always `48`
`anchor`	`{x, y}`	Compositing alignment point (see Body Part Specs)
`bounds`	`{x, y, w, h}`	Tight bounding box of all non-transparent pixels
`mutationZones`	array	Rectangles where procedural variants may modify pixels
`pixels`	array	Every non-transparent pixel: `{x, y, role}`

Pixel Format

Each pixel has: - `x`: horizontal position, 0 (left) to 47 (right) - `y`: vertical position, 0 (top) to 47 (bottom) - `role`: palette role index, 0-6 (see next section)

Only non-transparent pixels are listed. Any coordinate not in the array is fully transparent.

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4. Palette Roles

You paint with 7 abstract roles instead of specific colors. The rendering system maps these to actual RGBA colors based on the lobster's class.

Role	Index	Name	Purpose	Usage Guide
Outline	`0`	`outline`	Darkest color, defines silhouette edges	1px border around entire shape. **Every template must have a complete outline.**
Primary Shadow	`1`	`primaryShadow`	Dark shade of primary	Bottom edges, underside, depth shading, shadow areas
Primary Base	`2`	`primaryBase`	Main body color	Largest area — the primary fill of the body part
Primary Highlight	`3`	`primaryHighlight`	Light shade of primary	Top edges, light-catching surfaces, specular highlights
Secondary Base	`4`	`secondaryBase`	Contrasting detail color	Markings, patterns, joints, segmentation lines
Secondary Highlight	`5`	`secondaryHighlight`	Light shade of secondary	Highlights on secondary features
Accent	`6`	`accent`	Pop color, eye-catching	Sparingly used — eyes, glowing tips, special markings (2-8 pixels typical)

Role Distribution Guidelines

A well-balanced template typically has this approximate distribution:

Role	% of pixels	Notes
Outline (0)	25-35%	Complete 1px border, critical for silhouette
Primary Shadow (1)	10-20%	Bottom/right edges, depth
Primary Base (2)	25-40%	Largest area, main fill

Role	% of pixels	Notes
Primary Highlight (3)	8-15%	Top/left edges, catching light
Secondary Base (4)	5-15%	Class-distinctive patterns/details
Secondary Highlight (5)	2-8%	Highlights on secondary areas
Accent (6)	1-3%	Very sparse, maximum 2-8 pixels

Shading Convention

Light source is from the **top-left**: **- Top and left edges** → highlight (role 3) **- Bottom and right edges** → shadow (role 1) **- Interior** → base (role 2) **- Inner details/markings** → secondary (roles 4, 5) **- Special points** → accent (role 6)

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5. Class Palettes

Each of the 10 classes has a distinct color scheme. Your template roles map to these colors:

0: Bulwark — Tank (Steel Blue / Slate Gray / Titanium White)

Role	Color	Hex	Visual
0 Outline	Dark navy	`#1A2533`	Near-black blue
1 Shadow	Dark steel	`#385E90`	Darkened steel blue
2 Base	Steel blue	`#4682B4`	Main body color
3 Highlight	Light steel	`#6BA0D0`	Brightened steel blue
4 Secondary	Slate gray	`#708090`	Gray markings
5 Sec. Highlight	Light slate	`#8898A7`	Lighter gray
6 Accent	Titanium white	`#E8E8E8`	Near-white accents

1: Mantis — Assassin (Jade Green / Black / Venomous Yellow)

Role	Color	Hex	Visual
0 Outline	Dark forest	`#002E1D`	Very dark green
1 Shadow	Dark jade	`#008656`	Darkened jade
2 Base	Jade green	`#00A86B`	Main body color
3 Highlight	Light jade	`#33C98F`	Brightened jade
4 Secondary	Black	`#1A1A1A`	Dark markings
5 Sec. Highlight	Dark gray	`#3D3D3D`	Subtle lightened black
6 Accent	Venomous yellow	`#FFFF00`	Bright yellow accents

2: Leviathan — Bruiser (Deep Navy / Bronze / Dark Iron)

Role	Color	Hex	Visual
0 Outline	Abyss blue	`#000030`	Nearly pure black-blue
1 Shadow	Dark navy	`#000066`	Darkened navy
2 Base	Deep navy	`#000080`	Main body color
3 Highlight	Medium blue	`#3333A6`	Brightened navy
4 Secondary	Bronze	`#CD7F32`	Warm metallic markings
5 Sec. Highlight	Light bronze	`#DAA04D`	Lighter bronze

Role	Color	Hex	Visual
6 Accent	Dark iron	`#4A4A4A`	Gray-metal accents

3: Tempest — Nuker (Electric Blue / White / Arc-flash Cyan)

Role	Color	Hex	Visual
0 Outline	Storm gray	`#1A3A3D`	Dark teal-gray
1 Shadow	Muted electric	`#64C7CC`	Darkened electric blue
2 Base	Electric blue	`#7DF9FF`	Bright, vibrant blue
3 Highlight	Ice blue	`#A4FBFF`	Near-white blue
4 Secondary	White	`#F0F8FF`	Almost pure white
5 Sec. Highlight	Pure white	`#F5FAFF`	Brightest white
6 Accent	Cyan	`#00FFFF`	Arc-flash cyan

4: Specter — Debuffer (Ghost Purple / Ethereal Teal / Spectral White)

Role	Color	Hex	Visual
0 Outline	Void purple	`#1E1840`	Very dark purple
1 Shadow	Dark purple	`#6253BE`	Darkened purple
2 Base	Ghost purple	`#7B68EE`	Medium slate blue
3 Highlight	Light purple	`#9F93F3`	Brightened purple
4 Secondary	Teal	`#008B8B`	Ethereal teal
5 Sec. Highlight	Light teal	`#33A9A9`	Lighter teal
6 Accent	Spectral white	`#E0E0FF`	Pale lavender

5: Sentinel — Support (Gold / Warm White / Pearl)

Role	Color	Hex	Visual
0 Outline	Dark gold	`#3D3200`	Near-black gold
1 Shadow	Dark gold	`#CCAC00`	Darkened gold
2 Base	Gold	`#FFD700`	Rich gold
3 Highlight	Bright gold	`#FFE54D`	Light gold
4 Secondary	Warm white	`#FFFAF0`	Floral white
5 Sec. Highlight	Pure warm	`#FFFCF5`	Near-white
6 Accent	Pearl	`#F0E6D0`	Pearlescent cream

6: Reaver — DPS (Crimson / Dark Red / Bone White)

Role	Color	Hex	Visual
0 Outline	Blood black	`#3D0510`	Very dark red-black
1 Shadow	Dark crimson	`#B01030`	Darkened crimson
2 Base	Crimson	`#DC143C`	Vivid red
3 Highlight	Bright crimson	`#E5506A`	Lightened crimson
4 Secondary	Dark red	`#8B0000`	Deep blood red
5 Sec. Highlight	Medium red	`#AD3333`	Lighter dark red
6 Accent	Bone white	`#F5F0E0`	Off-white

7: Abyss — Lifesteal (Void Black / Toxic Green / Bioluminescent)

Role	Color	Hex	Visual
0 Outline	True black	`#000000`	Pure black
1 Shadow	Near black	`#0A0A0A`	Barely distinguishable from outline
2 Base	Void black	`#0D0D0D`	Almost black (very dark)
3 Highlight	Dark gray	`#3D3D3D`	Subtle light on darkness
4 Secondary	Toxic green	`#39FF14`	Neon green
5 Sec. Highlight	Light toxic	`#6BFF4E`	Bright neon green
6 Accent	Bioluminescent	`#00FF80`	Green glow

Special note for Abyss: The primary colors are extremely dark. The visual interest comes almost entirely from the secondary (toxic green) and accent (bioluminescent). Use more secondary/accent pixels than other classes — secondary markings should be prominent.

8: Kraken — Controller (Dark Teal / Bio-blue / Ink Black)

Role	Color	Hex	Visual
0 Outline	Deep teal	`#002020`	Very dark teal
1 Shadow	Dark teal	`#006666`	Darkened teal
2 Base	Teal	`#008080`	Classic teal
3 Highlight	Light teal	`#33A6A6`	Brightened teal
4 Secondary	Bio-blue	`#00BFFF`	Bright deep sky blue
5 Sec. Highlight	Light bio	`#33CFFF`	Lighter sky blue
6 Accent	Ink black	`#0A0A0A`	Near-black ink

9: Ember — Glass Cannon (Orange / Molten Red / Magma Yellow)

Role	Color	Hex	Visual
0 Outline	Burnt umber	`#3D1A00`	Dark brown-black
1 Shadow	Dark orange	`#CC5200`	Darkened orange
2 Base	Orange	`#FF6600`	Vivid orange
3 Highlight	Light orange	`#FF8F40`	Brightened orange
4 Secondary	Molten red	`#FF2400`	Scarlet red
5 Sec. Highlight	Light scarlet	`#FF5533`	Lighter red
6 Accent	Magma yellow	`#FFFF00`	Bright yellow

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6. Body Part Specifications

Each body part has a fixed **anchor point** for compositing alignment. All 6 layers are drawn on the same 48×48 canvas. The anchor is the reference point where the body part "attaches" to the lobster.

Compositing Layer Order (back to front)

```
Layer 1 (back):   Carapace  - main shell (largest, background)
Layer 2:          Legs      - walking legs (behind body)
Layer 3:          Tail      - tail fan (behind body)
```

```
Layer 4:          Eyes      - eye stalks (on top of shell)
Layer 5:          Antennae  - feelers (on top of shell)
Layer 6 (front):  Claws     - pincers (foreground, most visible)
```

Part 0: Carapace (Shell)

Property	Value
Anchor	`{x: 24, y: 20}` — center-back of shell
Stat affinity	HP (health)
Typical bounds	~24×18 pixels
Typical pixel count	350-450

Design direction: The main body shell. Dome-shaped, widest part of the lobster. This is the **largest single template** — occupies the center of the canvas.

Class differentiation: - **Bulwark**: Wide, thick, heavy-plated dome with reinforced ridges - **Mantis**: Sleek, narrow, streamlined shell with sharp edges - **Leviathan**: Massive, broad shell with ancient/barnacled texture - **Tempest**: Angular, crystalline shell with lightning-bolt edges - **Specter**: Ethereal, partially transparent-looking, wispy edges - **Sentinel**: Ornate, shield-like shell with decorative patterns - **Reaver**: Spiky, aggressive shell with barb-like protrusions - **Abyss**: Smooth, featureless void with toxic green vein patterns - **Kraken**: Organic, tentacle-ridge textures, deep-sea feel - **Ember**: Cracked/magma-veined shell, jagged heat-warped edges

Secondary/accent usage: Secondary color for shell patterns, segmentation lines, and class-specific markings. Accent for 2-4 special feature pixels (eyes of patterns, gem-like highlights).

Part 1: Claws (Pincers)

Property	Value
Anchor	`{x: 12, y: 28}` — front-center attachment
Stat affinity	Attack
Typical bounds	~23×22 pixels
Typical pixel count	300-400

Design direction: The signature fighting appendages. Drawn as a **pair** — both left and right claw in one template. Claws are the **frontmost layer** and most visually prominent feature.

Class differentiation: - **Bulwark**: Broad, shield-like crusher claws — defense-oriented shape - **Mantis**: Long, thin, blade-like raptorial claws — assassin strikes - **Leviathan**: Massive, heavy crusher claws — brute power - **Tempest**: Forked, trident-like claws — lightning rod shapes - **Specter**: Thin, ghostly, translucent-looking claws — wispy - **Sentinel**: Ornate, golden-rimmed claws — ceremonial appearance - **Reaver**: Serrated, saw-tooth claws — designed to rend - **Abyss**: Smooth claws with glowing vein lines — draining - **Kraken**: Tentacle-like gripping claws — wrapping, binding - **Ember**: Incandescent claws, flame-shaped tips — burning

Note: The claws template should include BOTH claws as a single unit. Position the pair symmetrically around the anchor point. The left claw is typically at x=2-12, the right claw "arm" extends to x=14-24.

Part 2: Tail

Property	Value
Anchor	`{x: 34, y: 36}` — back-low attachment
Stat affinity	Speed
Typical bounds	~21×21 pixels
Typical pixel count	250-350

Design direction: The tail fan and uropods. Extends from the back-bottom of the lobster. Shape conveys speed class identity.

Class differentiation: - **Bulwark**: Wide, flat, armored tail fan — stable, heavy - **Mantis**: Narrow, streamlined tail — aerodynamic/aquadynamic - **Leviathan**: Broad, powerful tail — rudder-like, massive - **Tempest**: Forked, lightning-bolt tail — split energy streams - **Specter**: Wispy, fading-edge tail — partially dissolved - **Sentinel**: Fan-shaped, ceremonial tail — peacock-like spread - **Reaver**: Barbed, weaponized tail — scorpion-sting vibe - **Abyss**: Tendril-like, dark with green veins — organic horror - **Kraken**: Multi-tipped tentacle tail — deep-sea appendage - **Ember**: Flame-shaped tail fan — rising fire pattern (see reference: `tail/ember.json`)

Note: The Ember reference template shows a multi-flame fan pattern with 5 flame tips at the top, converging to a narrow stalk at the bottom. Use this as a guide for the level of detail.

Part 3: Antennae

Property	Value
Anchor	`{x: 18, y: 10}` — top-front attachment
Stat affinity	Critical (crit chance)
Typical bounds	~20×16 pixels
Typical pixel count	150-250

Design direction: Two antennae/feelers extending from the top of the head. Thinner and more delicate than other parts. The most "fragile" looking part.

Class differentiation: - **Bulwark**: Short, sturdy, armored antennae — thick base, blunt tips - **Mantis**: Long, thin, whip-like antennae — sensitive, lethal precision - **Leviathan**: Thick, barnacle-encrusted antennae — ancient - **Tempest**: Crackling, forked antenna tips — lightning rods - **Specter**: Ghostly, semi-transparent, flickering antennae - **Sentinel**: Ornate, banner-like antennae — regal - **Reaver**: Barbed, hooked antennae — weaponized sensing - **Abyss**: Bioluminescent-tipped dark antennae — deep-sea lures - **Kraken**: Sucker-lined, tentacle-like antennae — grasping - **Ember**: Flame-tipped antennae — ember glow at tips

Note: Antennae are the **thinnest** body part. Most of the shape is 1-2 pixels wide. Use outline (0) for the main structure, with small clusters of primary/secondary at bases and tips. Accent pixels at the very tips for class flavor.

Part 4: Eyes

Property	Value
Anchor	`{x: 16, y: 18}` — front-high attachment
Stat affinity	Armor (defense)
Typical bounds	~16×14 pixels
Typical pixel count	150-250

Design direction: Two eyes on stalks, lobster-style. The stalks emerge from the front-top of the shell. Eyes are a key personality/expression feature.

Class differentiation: - **Bulwark**: Thick-stalked, armored eyes — visor-slit pupils - **Mantis**: Wide, compound-eye style — predator vision - **Leviathan**: Deep-set, ancient eyes — wise/heavy-lidded - **Tempest**: Bright, electric-glowing eyes — crackling energy - **Specter**: Hollow, ghostly eyes — empty glow - **Sentinel**: Bright, alert, golden-rimmed eyes — watchful guardian - **Reaver**: Narrow, aggressive, red-glowing eyes — predatory - **Abyss**: Void eyes with green pinpoint pupils — unsettling - **Kraken**: Large, intelligent, deep-sea eyes — observing - **Ember**: Ember-glowing, smoldering eyes — intense heat

Note: Eyes are paired (left and right stalk). The eye surfaces themselves are great places for accent (role 6) pixels — 2-4 accent pixels for iris/pupil highlights make the lobster feel alive.

Part 5: Legs

Property	Value
Anchor	`{x: 24, y: 38}` — bottom-center attachment
Stat affinity	HP (health, same as carapace)
Typical bounds	~28×12 pixels
Typical pixel count	200-300

Design direction: Walking legs (pereopods). Typically 3-4 pairs visible from a side/top-down perspective. Legs extend below and slightly to the sides of the body.

Class differentiation: - **Bulwark**: Thick, armored, sturdy legs — pillars - **Mantis**: Long, thin, articulated legs — fast movement - **Leviathan**: Massive, powerful legs — trunk-like - **Tempest**: Jointed, angular legs — lightning-bolt segments - **Specter**: Thin, fading legs — some legs seem to disappear - **Sentinel**: Armored, decorated legs — greaves/shin-guard look - **Reaver**: Spiked, aggressive legs — caltrops on joints - **Abyss**: Skeletal, dark legs with green joint markings - **Kraken**: Tentacle-hybrid legs — organic, flowing curves - **Ember**: Heat-cracked legs with glowing joint points

Note: Legs are drawn behind the main body (Layer 2, after carapace). They should peek out from under/around the shell. Use primarily outline and primary shadow/base colors — legs are less detailed than claws or carapace.

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7. Mutation Zones

Each template must define **2-4 rectangular mutation zones** — areas where the procedural variant system is allowed to modify pixels. Pixels outside these zones are **never changed** by the variant generator.

Purpose

The system generates 16 variants from each template by applying random mutations inside these zones:

Variant Range	Tier	Mutations Applied	Style
0	—	None (exact base template)	Original
1-3	Simple	0-2 mutations	Simplified, minor edge changes
4-7	Moderate	2-4 mutations	Small additions, thickening
8-11	Detailed	3-6 mutations	Patterns, overlays
12-15	Elaborate	5-8 mutations	Maximum elaboration

6 Mutation Types

Mutation	Description	Best Zone Placement
`add_pixels`	Extends silhouette at edges (spikes, bumps)	Outer edges, tips
`remove_pixels`	Thins/simplifies by removing non-outline pixels	Dense interior areas
`shift_pixels`	Moves a sub-region 1-2px for organic feel	Any interior area
`thicken`	Expands outlines inward	Thin areas, edges
`pattern_fill`	Adds interior patterns (dots, stripes, checkerboard)	Large flat interior areas
`detail_overlay`	Places small 2×2 or 3×3 decorative clusters	Mid-body feature areas

Zone Placement Guidelines

1. **Zone 1 — Edge zone**: Place along the most distinctive silhouette edge (top of shell, claw tips, tail fan edge). Allow `shift_pixels`, `thicken`, `add_pixels`.
2. **Zone 2 — Interior zone**: Place over the largest flat interior area. Allow `pattern_fill`, `detail_overlay`, `add_pixels`.
3. **Zone 3 — Detail zone**: Place over a secondary feature area (joint, marking, pattern). Allow `shift_pixels`, `add_pixels`, `remove_pixels`.
4. **Optional Zone 4**: Only if the part is large enough. Cover a distinct sub-feature.

Rules

- Zones **may overlap** — the system handles this correctly
- Zones must stay **within the 48×48 canvas** (`x + w <= 48`, `y + h <= 48`)
- Zones should **not cover the entire template** — leave some areas stable
- **Do not place zones over critical structural pixels** (like the single anchor attachment point area) unless you want them to vary
- Each zone must list **at least 1** allowed mutation type, recommend **2-3**
- Zones covering small areas (< 4×4) should only allow `shift_pixels` or `add_pixels`

Example (from `carapace/bulwark.json`)

```
"mutationZones": [
  {
    "x": 14, "y": 13, "w": 20, "h": 4,
    "allowed": ["shift_pixels", "thicken", "detail_overlay"]
  },
  {
    "x": 13, "y": 17, "w": 22, "h": 6,
    "allowed": ["pattern_fill", "add_pixels", "detail_overlay"]
  },
  {
    "x": 15, "y": 23, "w": 18, "h": 5,
    "allowed": ["add_pixels", "remove_pixels", "thicken"]
  }
]
```

This places zones across: the top highlight band (zone 1), the middle body (zone 2), and the lower edge (zone 3). The shell's outline border pixels and anchor area are outside the zones.

8. Asset Checklist (60 Templates)

Completed (3/60) — Use as Reference

#	Body Part	Class	File	Pixel Count
1	Carapace	Bulwark (0)	`carapace/bulwark.json`	~364
2	Claws	Mantis (1)	`claws/mantis.json`	~340
3	Tail	Ember (9)	`tail/ember.json`	~350

Remaining (57/60) — To Be Created

Carapace (9 remaining)

#	Class	`classAffinity`	File	Notes
4	Mantis	1	`carapace/mantis.json`	Sleek, streamlined shell
5	Leviathan	2	`carapace/leviathan.json`	Massive, ancient shell
6	Tempest	3	`carapace/tempest.json`	Angular, crystalline shell
7	Specter	4	`carapace/specter.json`	Ethereal, wispy-edged shell
8	Sentinel	5	`carapace/sentinel.json`	Ornate shield-like shell
9	Reaver	6	`carapace/reaver.json`	Spiky, aggressive shell
10	Abyss	7	`carapace/abyss.json`	Smooth void + green veins
11	Kraken	8	`carapace/kraken.json`	Organic, tentacle-ridged shell
12	Ember	9	`carapace/ember.json`	Cracked/magma-veined shell

Claws (9 remaining)

#	Class	`classAffinity`	File	Notes
13	Bulwark	0	`claws/bulwark.json`	Broad shield-crusher claws
14	Leviathan	2	`claws/leviathan.json`	Heavy crusher claws
15	Tempest	3	`claws/tempest.json`	Forked trident-like claws
16	Specter	4	`claws/specter.json`	Ghostly thin claws

#	Class	`classAffinity`	File	Notes
17	Sentinel	5	`claws/sentinel.json`	Golden ornate claws
18	Reaver	6	`claws/reaver.json`	Serrated saw-tooth claws
19	Abyss	7	`claws/abyss.json`	Dark claws w/ green veins
20	Kraken	8	`claws/kraken.json`	Tentacle-gripping claws
21	Ember	9	`claws/ember.json`	Flame-tipped incandescent claws

Tail (9 remaining)

#	Class	`classAffinity`	File	Notes
22	Bulwark	0	`tail/bulwark.json`	Wide armored tail fan
23	Mantis	1	`tail/mantis.json`	Narrow streamlined tail
24	Leviathan	2	`tail/leviathan.json`	Broad powerful tail
25	Tempest	3	`tail/tempest.json`	Forked lightning tail
26	Specter	4	`tail/specter.json`	Wispy fading-edge tail
27	Sentinel	5	`tail/sentinel.json`	Fan-shaped ceremonial tail
28	Reaver	6	`tail/reaver.json`	Barbed scorpion-sting tail
29	Abyss	7	`tail/abyss.json`	Tendril dark tail
30	Kraken	8	`tail/kraken.json`	Multi-tip tentacle tail

Antennae (10 — all needed)

#	Class	`classAffinity`	File	Notes
31	Bulwark	0	`antennae/bulwark.json`	Short, sturdy, armored
32	Mantis	1	`antennae/mantis.json`	Long, thin, whip-like
33	Leviathan	2	`antennae/leviathan.json`	Thick, barnacle-encrusted
34	Tempest	3	`antennae/tempest.json`	Crackling forked tips
35	Specter	4	`antennae/specter.json`	Ghostly, flickering
36	Sentinel	5	`antennae/sentinel.json`	Ornate, banner-like

#	Class	`classAffinity`	File	Notes
37	Reaver	6	`antennae/reaver.json`	Barbed, hooked
38	Abyss	7	`antennae/abyss.json`	Bioluminescent-tipped
39	Kraken	8	`antennae/kraken.json`	Suction-lined tentacle
40	Ember	9	`antennae/ember.json`	Flame-tipped

Eyes (10 — all needed)

#	Class	`classAffinity`	File	Notes
41	Bulwark	0	`eyes/bulwark.json`	Thick-stalked, armored
42	Mantis	1	`eyes/mantis.json`	Wide compound-eye style
43	Leviathan	2	`eyes/leviathan.json`	Deep-set, ancient
44	Tempest	3	`eyes/tempest.json`	Electric-glowing
45	Specter	4	`eyes/specter.json`	Hollow, ghostly
46	Sentinel	5	`eyes/sentinel.json`	Bright, golden-rimmed
47	Reaver	6	`eyes/reaver.json`	Narrow, aggressive
48	Abyss	7	`eyes/abyss.json`	Void with green pinpoints
49	Kraken	8	`eyes/kraken.json`	Large, intelligent
50	Ember	9	`eyes/ember.json`	Smoldering, intense

Legs (10 — all needed)

#	Class	`classAffinity`	File	Notes
51	Bulwark	0	`legs/bulwark.json`	Thick, armored pillars
52	Mantis	1	`legs/mantis.json`	Long, thin, articulated
53	Leviathan	2	`legs/leviathan.json`	Massive, trunk-like
54	Tempest	3	`legs/tempest.json`	Angular, lightning-bolt
55	Specter	4	`legs/specter.json`	Thin, fading
56	Sentinel	5	`legs/sentinel.json`	Armored greaves
57	Reaver	6	`legs/reaver.json`	Spiked, aggressive
58	Abyss	7	`legs/abyss.json`	Skeletal, green joints
59	Kraken	8	`legs/kraken.json`	Tentacle-hybrid curves
60	Ember	9	`legs/ember.json`	Heat-cracked, glowing joints

---

9. Template Editor Tool

A browser-based pixel editor is included at:

`packages/asset-gen/tools/template-editor.html`

Open it directly in any browser (no build step needed).

Features

- **48×48 grid** at 12× zoom with toggleable grid lines
- **7-role palette bar** — click a role to select, then click pixels to paint
- **Eraser tool** — right-click to erase pixels (set to transparent)

- **Live preview strip** — shows the template rendered in all 10 class palettes simultaneously
- **Mutation zone drawing** — shift+drag to define rectangular mutation zones
- **JSON export** — click Export to get the complete template JSON
- **JSON import** — paste existing template JSON and click Import to continue editing
- **Keyboard shortcuts**:
  - `0-6` — select role 0-6
  - `E` — select eraser
  - `G` — toggle grid
  - `Z` — toggle mutation zone mode

Workflow

1. Open `template-editor.html` in browser
2. Select a role from the palette bar (start with role 0 for outline)
3. Draw the body part outline first, then fill interior with roles 1-6
4. Switch to mutation zone mode (Z key) and draw 2-4 zones
5. Check the live preview strip — all 10 classes should look good
6. Export JSON → save to `packages/asset-gen/src/templates/data/<part>/<class>.json`
7. Validate: `bun packages/asset-gen/src/cli.ts validate-templates`

Validation CLI Commands

```
# Validate all templates
bun packages/asset-gen/src/cli.ts validate-templates

# Render a single variant to check quality
bun packages/asset-gen/src/cli.ts variant carapace bulwark 0 --size 192

# Generate a 16-variant sheet for visual QA
bun packages/asset-gen/src/cli.ts sheet carapace bulwark --size 96

# Preview all templates (grid of everything)
bun packages/asset-gen/src/cli.ts preview-all --cellsize 48
```

---

10. Quality Criteria & Validation

Automatic Validation (run `validate-templates`)

The system checks: - `version` is `1` - `width` and `height` are both `48` - `bodyPart` is one of the 6 valid names - `classAffinity` is 0-9 - `anchor` has numeric `x` and `y` - `bounds` has numeric `x`, `y`, `w`, `h` - All mutation zones are within canvas bounds (0-47) - Each mutation zone has at least 1 valid `allowed` mutation type - All pixels have valid `x` (0-47), `y` (0-47), and `role` (0-6) - No duplicate pixel coordinates

Manual Quality Checks

After creating each template, verify:

1. **Complete outline**: Every non-transparent pixel that borders a transparent pixel should have an outline-adjacent pixel (role 0). The silhouette must be fully enclosed.

2. **Correct anchor point**: Must match the default anchors listed in Body Part Specs. This is critical for compositing — wrong anchors cause parts to misalign.

3. **Accurate bounds**: The `bounds` rectangle must tightly contain all pixels. Calculate as: `{x: minX, y: minY, w: maxX-minX+1, h: maxY-minY+1}`.

4. **Cross-class compositing**: Render your template with parts from OTHER classes. A Bulwark carapace must look good paired with a Mantis claw and an Ember tail. No pixel overlap conflicts between layers.

5. **Variant quality**: Generate all 16 variants (`sheet` command) and verify:

   - Variant 0 matches the base template exactly
   - No variants produce disconnected pixels (floating dots)
   - No variants break the outline
   - Elaborate variants (12-15) look interestingly different from the base

6. **Pixel count range**: Templates should have 150-450 pixels depending on body part:

Body Part	Min Pixels	Max Pixels	Typical
Carapace	300	500	350-450
Claws	250	450	300-400
Tail	200	400	250-350
Antennae	100	250	150-250
Eyes	100	250	150-250
Legs	150	350	200-300

7. **Role balance**: Check the approximate role distribution matches the guidelines in Section 4. Avoid templates that are 90% one role.

8. **10-class preview**: The template should look visually coherent across all 10 class palettes. The live preview strip in the editor shows this.

---

11. File Naming & Folder Structure

```
packages/asset-gen/src/templates/data/
  carapace/
     bulwark.json     ← classAffinity: 0
     mantis.json      ← classAffinity: 1
     leviathan.json   ← classAffinity: 2
     tempest.json     ← classAffinity: 3
     specter.json     ← classAffinity: 4
     sentinel.json    ← classAffinity: 5
     reaver.json      ← classAffinity: 6
     abyss.json       ← classAffinity: 7
     kraken.json      ← classAffinity: 8
     ember.json       ← classAffinity: 9
  claws/
     (same 10 files)
  tail/
     (same 10 files)
  antennae/
     (same 10 files)
  eyes/
     (same 10 files)
  legs/
     (same 10 files)
```

File naming: `<class_name>.json` (lowercase, matching the class names exactly) **Folder naming**: `<body_part_name>/` (lowercase, matching the body part names exactly)

The loader resolves templates by path: `data/<bodyPartName>/<className>.json`

---

12. Reference Examples

Example 1: `carapace/bulwark.json` (Shell — Tank)

Key features to study: - 364 pixels spanning y=11 to y=28 (18 rows) - Complete outline border (role 0) around entire dome shape - Top rows use highlight (3) for light-catching edge - Middle rows use primary base (2) as main fill - Secondary base (4) and secondary highlight (5) form horizontal stripe pattern through middle - 4 accent pixels (6) placed symmetrically for decorative detail - Bottom rows transition to shadow (1) - Anchor at exact center: {24, 20} - 3 mutation zones: top band, middle body, lower edge

Example 2: `claws/mantis.json` (Pincers — Assassin)

Key features to study: - ~340 pixels forming two blade-like shapes - Main claw body (upper-left area, x=2-13) with a secondary arm/pincer (lower-right, x=14-24) - Small "finger" protrusion at bottom (x=10-13, y=33-37) - Secondary (4,5) colors used for inner claw detail and joint accent pattern - Accent (6) used sparingly for glow spots (2-3 pixels) - Anchor at {12, 28} — front attachment

Example 3: `tail/ember.json` (Tail — Glass Cannon)

Key features to study: - ~350 pixels forming a 5-flame fan pattern - 5 individual flame tips at y=18-22 (upper portion) - Flames merge into a single stalk at y=26-38 - Secondary (4,5) colors create a "spine" pattern down the center - Accent (6) marks the hottest points (3 pixels) - Narrow stalk tapers from ~10px wide to ~2px wide at base - Anchor at {34, 36} — back attachment - 3 mutation zones: flame tips, mid-body spine, stalk

---

13. Visual Style Guide

Overall Aesthetic

- **Pixel art, not pixel perfect**: Lobsters should feel organic and characterful, not geometric
- **Chunky proportions**: At 48×48, every pixel matters. Favor readability over realism
- **Class identity over anatomical accuracy**: A Mantis claw should scream "assassin" more than "biologically accurate pincer"
- **Silhouette-first design**: The outline (role 0) is the most important layer. A good template is recognizable from its silhouette alone

Do's

- Start every template by drawing the outline first
- Use 1px outlines consistently (don't go 2px thick)
- Place highlights on top-left edges, shadows on bottom-right
- Use accent sparingly — it's the "pop" color, 2-8 pixels max
- Make each class version visually distinct from the same body part in other classes
- Test cross-class compositing regularly (e.g., Bulwark shell + Ember tail)
- Keep mutation zones away from the anchor attachment area

Don'ts

- Don't leave gaps in the outline — the silhouette must be airtight
- Don't use only 1-2 roles — use at least 5 of 7 for visual richness
- Don't make templates too small (< 100 pixels) — they'll look empty at display size

- Don't make templates too large (> 500 pixels) — the canvas only has 2304 total pixels
- Don't place pixels at the extreme canvas edges (x=0, x=47, y=0, y=47) — leave 1-2px margin for mutation zones to expand into
- Don't forget to set the correct `classAffinity` value matching the class index (0-9)
- Don't copy exact shapes between classes — the whole point is class identity through distinct shapes

Priority Order for Creation

Recommended order (prioritize completing full body part sets):

1. **Carapace** (9 remaining) — the backbone of every lobster
2. **Claws** (9 remaining) — the most visually prominent feature
3. **Eyes** (10) — critical for personality
4. **Tail** (9 remaining) — key shape differentiator
5. **Legs** (10) — background element, can be simpler
6. **Antennae** (10) — thinnest/simplest part

This order lets you render progressively more complete lobsters as you go — a carapace + claws already looks like a recognizable lobster.

Summary

Deliverable	Count	Format	Location
Template JSON files	57 remaining (60 total)	JSON	packages/asset-gen/src/templates/dat
Canvas size	—	48×48 pixels	—
Palette roles	7 per template	Role indices 0-6	—
Mutation zones	2-4 per template	Rectangles with allowed mutations	Inside each JSON
Validation	Per template	bun packages/asset-gen/src/cli.ts validate-templates	—
QA sheets	Per template	bun packages/asset-gen/src/cli.ts sheet <part> <class>	—